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L^2 estimates for the Solutions of Burgers-Sivashinsky Equation in 1D and 2D

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Abstract. In this paper, in the first part, we work on the solutions of one- dimensional Burgers-Sivashinsky Equation in the bounded domain $[-L,L]$. In the second part, we consider radially symmetric solutions of this equation in two and higher dimension in a bounded domain $[0,R]$. In both cases, using Lyapunov function approach, we study the long time behavior of the solutions and prove that there exists a time independent bound for the L^2 norm of the solutions. Thus in each case there exists an absorbing ball when time tends to infinity.

Keywords: Burgers-Sivashinsky Equation, L^2 bounds

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INTRODUCTION

Burgers-Sivashinsky Equation,

$$\phi_t = \Delta\phi + \phi - |\nabla\phi|^2 \quad (1)$$

is often used as a model problem for fluid dynamical systems. In the case of one space dimension, when we differentiate (1) and define $u = \phi_x$, we have

$$u_t = u_{xx} + u - 2uu_x \quad (2)$$

We will work with this equation assuming first that the initial data is odd, then we will generalize the result to general initial data. We will have the following boundary conditions and will assume that u is the mean-zero solution, i.e.,

$$\frac{d^j u}{dx^j}(L) = \frac{d^j u}{dx^j}(-L), \quad j = 0, 1 \quad \int_{-L}^L u_0(x) dx = 0, \quad x \in (-L, L) \quad (3)$$

Note that throughout this paper, we use $\|\cdot\|_2$ to denote $\|\cdot\|_{L^2[-L,L]}$. We also use $\bar{H}^1[-L, L]$ to denote the Sobolev Space obtained by taking the completion with respect to the norm $\|\cdot\|_{H^1}$ of smooth functions satisfying the boundary conditions $\phi(-L) = \phi(L)$. A dot above any space will denote the subspace of the functions of zero mean, i.e $\phi \in \dot{H}^1[-L, L]$ if and only if $\phi \in \bar{H}^1[-L, L]$ and $\int_{-L}^L \phi(x) dx = 0$.

Fundamental Lemmas

Lemma 0.1 Given $u = u(x, t) \in L^2[-L, L]$ and $\phi(x, t) = \phi(x) \in L^2[-L, L]$ satisfying the following inequality:

$$\frac{d}{dt} \|u - \phi\|_2^2 \leq -\lambda_0 \|u\|_2^2 + P^2 \quad (4)$$

for some constants $\lambda_0 > 0$ and P , then $B(O, R^{**})$, the ball of radius R^{**} centered about the origin, is an attracting region, where the radius R^{**} is given by

$$R^{**} = \sqrt{2\|\phi\|_2^2 + \frac{2P^2}{\lambda_0} + \|\phi\|_2^2} \quad (5)$$

Proof. The proof is in [Bronschi and Gambill(2006)].

Lemma 0.2 For any $\phi \in \dot{H}^1[-L, L]$ and a solution of (2), $u(t; x)$ we have the inequality

$$\frac{d}{dt} \int_{-L}^L (u - \phi)^2 dx \leq 2 \int_{-L}^L (2 - \phi_x) u^2 - \frac{3}{4} u_x^2 dx + \int_{-L}^L \frac{1}{2} \phi^2 + 2\phi_x^2 dx \quad (6)$$

A straightforward calculation gives

$$\frac{1}{2} \frac{d}{dt} \|u - \phi\|^2 = \int_{-L}^L u_t (u - \phi) dx = \int_{-L}^L (u_{xx} + u - 2uu_x)(u - \phi) dx \quad (7)$$

After integrating by parts and applying periodic boundary conditions, we have

$$\frac{1}{2} \frac{d}{dt} \|u - \phi\|^2 = \int_{-L}^L -u_x^2 + u^2 + u_x \phi_x - u \phi - u^2 \phi_x dx \quad (8)$$

Applying the Cauchy-Schwartz inequality in the form $\langle f, g \rangle \leq p/2 \langle f, f \rangle + 1/2p \langle g, g \rangle$ gives

$$\frac{1}{2} \frac{d}{dt} \|u - \phi\|^2 \leq \int_{-L}^L \left(1 + \frac{1}{2p} - \phi_x\right) u^2 + \left(-1 + \frac{1}{2q}\right) u_x^2 + \frac{p}{2} \phi^2 + \frac{q}{2} \phi_x^2 dx \quad (9)$$

Taking $p = \frac{1}{2}$ and $q = 2$ we get

$$\frac{1}{2} \frac{d}{dt} \|u - \phi\|^2 \leq \int_{-L}^L (2 - \phi_x) u^2 - \frac{3}{4} u_x^2 dx + \int_{-L}^L \frac{1}{4} \phi^2 + \phi_x^2 dx \quad (10)$$

Note that (4) and (6) show that if one can construct $\phi \in \dot{H}^1[-L, L]$ such that the coercivity estimate

$$\langle u, Ku \rangle = 2 \int_{-L}^L \left(\frac{3}{4} u_x^2 + (\phi_x - 2) u^2 \right) dx \geq \lambda_0 \|u\|_2^2 > 0 \quad (11)$$

holds for some λ_0 independent of L , then one gets an estimate of the form

$$\limsup_{t \rightarrow \infty} \|u\|_2 \leq R^{**} = \sqrt{c_1 \|\phi\|_2^2 + c_2 \|\phi_x\|_2^2} + \|\phi\|_2 \leq C \|\phi\|_{\tilde{H}^1} < \infty$$

In the following section, defining $\lambda = \frac{\lambda_0}{2}$, we will construct $\phi \in \dot{H}^1[-L, L]$ such that

$$\int_{-L}^L \frac{3}{4} u_x^2 + (\phi_x - 2) u^2 dx \geq \frac{\lambda_0}{2} \|u\|^2 = \lambda \|u\|^2 \quad (12)$$

Construction of ϕ

We construct the mean zero function in this form: $\phi_x := q(x) - \langle q \rangle$. Here $\langle \cdot \rangle$ denotes the mean value on $[-L, L]$:

$$\langle q \rangle = \frac{1}{2L} \int_{-L}^L q(x) dx \quad (13)$$

We define an even function $\varphi \in C_0^\infty$ such that $\text{supp } \varphi \in [-4, -\frac{1}{4}] \cup [\frac{1}{4}, 4]$ and $\varphi(x) = 1$ if $|x| \in [\frac{1}{2}, 2]$ and $\int_{-L}^L \varphi(x) dx \leq 1$. We define the following potential function q_M as

$$q_M = -M^2 \varphi(Mx) \quad (14)$$

Note that that $\langle q_M(x) \rangle \leq -\frac{M}{2L}$ since

$$\int_{-L}^L q_M(x) dx = \int_{-L}^L -M^2 \varphi(Mx) dx = -M \int_{-L}^L \varphi(y) dy \quad (15)$$

Our aim is to choose q such that it satisfies

$$\checkmark \langle q \rangle \leq -2 - \lambda \quad (16)$$

$$\checkmark \int_{-L}^L \frac{3}{4} u_x^2 + q(x) u^2 dx \geq 0 \quad (17)$$

If we choose $q = \frac{q_M}{100}$, then (16) implies that there exists M such that $M \geq 200(2 + \lambda)L$. So for such M , we have to show (17). By Hardy's Inequality with the assumption of $u(0) = 0$, we get

$$\int_{-L}^L u_x^2 dx \geq \frac{1}{4} \int_{-L}^L \frac{u^2}{x^2} dx \quad (18)$$

Using (18), we get

$$\int_{-L}^L \frac{3}{4} u_x^2 + q(x) u^2 dx \geq \int_{-L}^L \frac{3}{16} \frac{u^2}{x^2} + q(x) u^2 dx = \int_{-L}^L \left(\frac{3}{16x^2} - \frac{M^2}{100} \varphi(Mx) \right) u^2 dx \quad (19)$$

Observe that if $|x|$ is not in $[\frac{1}{4M}, 4M]$, then (17) automatically satisfies. Otherwise

$$\int_{-L}^L \frac{3}{4} u_x^2 + q(x) u^2 dx \geq \int_{-L}^L M^2 \left(\frac{3}{256} - \frac{1}{100} \varphi(Mx) \right) u^2 dx \geq 0 \quad (20)$$

Thus we can conclude that

$$\int_{-L}^L \frac{3}{4} u_x^2 + (\phi_x - 2 - \lambda_0) u^2 dx \geq \int_{-L}^L \frac{3}{4} u_x^2 + q(x) u^2 dx \geq 0 \quad (21)$$

Note that in order to minimize $\|\phi\|_{\tilde{H}^1}$, we will choose $M = O(L)$ since $M \geq 200(2 + \lambda)L$. Thus an optimal potential function is in the form $q(x) = -L^2 \varphi(Lx)$.

Lemma 0.3 *The potential ϕ satisfies $\|\phi\|_{\tilde{H}^1} \leq CL^{3/2}$*

Proof. From the definition $\phi_x = q(x) - \langle q \rangle = -L^2 \varphi(Lx) - \langle q \rangle$

$$\|\phi_x\|_{L^2}^2 = \int_{-L}^L q^2(x) dx - \int_{-L}^L \langle q \rangle^2 dx = \int_{-L}^L L^4 \varphi^2(Lx) dx - \int_{-L}^L \langle q \rangle^2 dx$$

After rescaling we get $\|\phi_x\|_2 = O(L^{3/2})$. From the definition of ϕ_x , we have

$$\phi(x) = \int_0^x \phi_s ds = \int_0^x q(s) - \langle q \rangle ds = \int_0^x -L^2 \varphi(Ls) - \langle q \rangle ds \quad (22)$$

After the substitution $y = Ls$, we get $\phi(x) = -L \int_0^{Lx} \varphi dy - \langle q \rangle x$

Since φ is bounded and supported on $[-4, -\frac{1}{4}] \cup [\frac{1}{4}, 4]$ we have $|\phi(x)| \leq CL + \langle q \rangle L = O(L)$. Thus we have $\|\phi\|_2^2 \leq (2L) \|\phi\|_{L^\infty}^2 = O(L^3)$.

Extension to arbitrary initial data

In order to extend our claims to any arbitrary initial data, we will use the idea introduced in [Collet et al.(1993)]. We will define the potential function as $\phi_b(x) = \phi(x + b)$ where $b = b(t)$. The point b will be chosen in such a way that the gradient of the distance function is parallel to a line connecting u to the closest point on the curve, union of such comparison functions ϕ_b . We will also define $b(t)$ such that $b(0) = 0$ and satisfying $b'(t) = \frac{1}{2L(\lambda + 2)} \int_{-L}^L u \phi_b' dx$. Note that we have

$$-\int_{-L}^L u \phi_b' dx = \frac{1}{2} \partial_b \int_{-L}^L (u - \phi_b)^2 dx \quad \text{and} \quad \int_{-L}^L u \phi_b' dx = \int_{-L}^L (u - \phi) \phi_b' dx \quad (23)$$

Claim

$$\frac{1}{2} \frac{d}{dt} \|u - \phi_b\|_2^2 + \frac{1}{2L(\lambda + 2)} \left(\int_{-L}^L u \phi_b' dx \right)^2 \leq \int_{-L}^L (2 - \phi_b') u^2 - \frac{3}{4} u_x^2 + \phi^2 + \frac{1}{4} \phi_b'^2 dx \quad (24)$$

Proof.

$$\begin{aligned}\frac{1}{2} \frac{d}{dt} \|u - \phi_b\|_2^2 &= \frac{1}{2} \frac{d}{dt} \int_{-L}^L (u - \phi_b)^2 dx = \int_{-L}^L (u - \phi_b) u_t dx - \int_{-L}^L (u - \phi_b) \frac{\partial}{\partial b} \phi_b b'(t) dx \\ &= \int_{-L}^L (u - \phi_b) u_t dx - b'(t) \int_{-L}^L u \phi_b' dx\end{aligned}$$

Thus we get $\frac{1}{2} \frac{d}{dt} \|u - \phi_b\|_2^2 + \frac{1}{2L(\lambda + 2)} \left(\int_{-L}^L u \phi_b' dx \right)^2 =$

$$\int_{-L}^L (u - \phi_b) (u_{xx} + u - 2uu_x) dx - b'(t) \int_{-L}^L u \phi_b' dx + \frac{1}{2L(\lambda + 2)} \left(\int_{-L}^L u \phi_b' dx \right)^2$$

Since we choose $b'(t) = \frac{1}{2L(\lambda + 2)} \left(\int_{-L}^L u \phi_b' dx \right)$, then the two last terms cancel and we get

$$\frac{1}{2} \frac{d}{dt} \|u - \phi_b\|_2^2 + \frac{1}{2L(\lambda + 2)} \left(\int_{-L}^L u \phi_b' dx \right)^2 = \int_{-L}^L (u - \phi_b) (u_{xx} + u - 2uu_x) dx$$

As in the case of odd initial data, because of (6), the claim follows.

Construction of ϕ_b

Similar to the idea in [Collet et al.(1993)], our aim will be to construct the potential function ϕ_b such that we have

$$\int_{-L}^L \frac{3}{4} u_x^2 + (\phi_b' - 2) u^2 dx + \frac{1}{2L(\lambda + 2)} \left(\int_{-L}^L u \phi_b' dx \right)^2 \geq \frac{\lambda_0}{2} \|u\|_2^2 = \lambda \|u\|_2^2 \quad (25)$$

which is equivalent to

$$\int_{-L}^L \frac{3}{4} u_x^2 + (\phi_b' - \lambda - 2) u^2 dx + \frac{1}{2L(\lambda + 2)} \left(\int_{-L}^L u \phi_b' dx \right)^2 \geq 0 \quad (26)$$

It suffices to prove the claim for $b = 0$ since (26) is invariant under translation. So we will write $\phi_{b=0} = \phi$ and $u = u(0) + u_a + u_s$ where u_a is the antisymmetric part and u_s is the symmetric part of u .

First, observe that since $\int_{-L}^L u_a' u_s' dx = 0$, then $\int_{-L}^L u_x^2 dx = \int_{-L}^L u_a'^2 + u_s'^2 dx$.

$$\begin{aligned}\text{Second observe that } \int_{-L}^L (\phi_x - \lambda - 2) u^2 dx &= \int_{-L}^L (\phi_x - \lambda - 2) (u(0) + u_s + u_a)^2 dx \\ &= \int_{-L}^L (\phi_x - \lambda - 2) (u_s^2 + u_a^2) dx + \int_{-L}^L (\phi_x - \lambda - 2) u(0)^2 dx + \\ &2 \int_{-L}^L (\phi_x - \lambda - 2) u(0) u_s dx + 2 \int_{-L}^L (\phi_x - \lambda - 2) u(0) u_a dx + 2 \int_{-L}^L (\phi_x - \lambda - 2) u_a u_s dx\end{aligned}$$

Using the fact that ϕ_x is even and $\int_{-L}^L u_s dx = -2Lu(0)$, we get $\int_{-L}^L (\phi_x - \lambda - 2)u^2 dx =$

$$= \int_{-L}^L (\phi_x - \lambda - 2)(u_s^2 + u_a^2) dx + 2L(\lambda + 2)u(0)^2 + 2u(0) \int_{-L}^L \phi_x u dx$$

Thus we get $\int_{-L}^L \frac{3}{4}u_x^2 + (\phi_x - \lambda - 2)u^2 dx + \frac{1}{2L(\lambda + 2)} \left(\int_{-L}^L u \phi_x dx \right)^2 =$

$$= \int_{-L}^L \frac{3}{4}u_a'^2 + (\phi_x - \lambda - 2)u_a^2 dx + \int_{-L}^L \frac{3}{4}u_s'^2 + (\phi_x - \lambda - 2)u_s^2 dx + 2L(\lambda + 2)u(0)^2$$

$$+ 2u(0) \int_{-L}^L \phi_x u dx + \frac{1}{2L(\lambda + 2)} \left(\int_{-L}^L u \phi_x dx \right)^2$$

Since $u_s(0) = 0$ and $u_a(0) = 0$, we can construct ϕ such that the coercivity estimates for u_s and u_a hold. Thus we can have the following inequality which proves the coercivity estimate.

$$\int_{-L}^L \frac{3}{4}u_x^2 (\phi_x - \lambda - 2)u^2 dx + \frac{1}{2L(\lambda + 2)} \left(\int_{-L}^L u \phi_x dx \right)^2 = \int_{-L}^L \frac{3}{4}u_a'^2 + (\phi_x - \lambda - 2)u_a^2 dx$$

$$+ \int_{-L}^L \frac{3}{4}u_s'^2 + (\phi_x - \lambda - 2)u_s^2 dx + \left(\frac{1}{\sqrt{2L(\lambda + 2)}} \int_{-L}^L u \phi_x dx + \sqrt{2L(\lambda + 2)}u(0) \right)^2 \geq 0$$

L^2 BOUNDS FOR RADially SYMMETRIC SOLUTIONS OF THE BURGERS-SIVASHINSKY EQUATION IN DIMENSION TWO

In this part of the paper we will work on the radially symmetric solutions of 2D-Burgers-Sivashinsky equation on the domain $\Omega = \{x \in \mathbb{R}^2 : 0 \leq \|x\| < R_0\}$. Changing the rectangular coordinates to polar coordinates and defining the radially symmetric solution as $v(r)$, (1) becomes

$$v_t - \left(\frac{\partial^2 v}{\partial r^2} + \frac{1}{r} \frac{\partial v}{\partial r} \right) - v + \left(\frac{\partial v}{\partial r} \right)^2 = 0 \quad (27)$$

And we know that v is even due to its construction. Differentiating (27), and defining $u = \frac{\partial v}{\partial r}$ we have

$$u_t - \frac{\partial^2 u}{\partial r^2} - \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} u - u - 2u \frac{\partial u}{\partial r} = 0 \quad (28)$$

We will have boundary conditions with the assumption of $u(r)$ being the mean-zero solution, i.e.,

$$\frac{d^j u}{dr^j}(R) = \frac{d^j u}{dr^j}(0), \quad j = 0, 1, \quad \int_0^R u_0(r) dr = 0, \quad r \in (0, R) \quad (29)$$

We will work on the following equation, and the space will be $L^2[0, R]$ and we know that u is an odd function, due to its construction.

$$u_t - u_{rr} - \frac{1}{r}u_r + \frac{1}{r^2}u - u + 2uu_r = 0 \quad (30)$$

Note that throughout this section, we will define $\|\cdot\|_2 = \left(\int_0^R (\cdot)^2 dr\right)^{1/2}$.

Lemma 0.4 For any $\phi \in \dot{H}^1[0, R]$ and $u(r, t)$ solving (30), we have the inequality

$$\frac{1}{2} \frac{d}{dt} \|u - \phi\|^2 \leq \int_0^R -\frac{3}{4}u_r^2 + (2 - \phi_r)u^2 dr + \int_0^R 126\phi_r^2 + \frac{1}{4}\phi^2 dr \quad (31)$$

Proof. A straightforward calculation gives that

$$\frac{1}{2} \frac{d}{dt} \|u - \phi\|^2 = \int_0^R u_t(u - \phi) dr = \int_0^R (u_{rr} + \frac{1}{r}u_r - \frac{1}{r^2}u + u - 2uu_r)(u - \phi) dr \quad (32)$$

After integrating by parts and applying periodic boundary conditions, we have

$$\frac{1}{2} \frac{d}{dt} \|u - \phi\|^2 = \int_0^R -u_r^2 + \frac{1}{r}uu_r - \frac{1}{r^2}u^2 + u^2 + u_r\phi_r - \frac{1}{r}u_r\phi + \frac{1}{r^2}u\phi - u\phi - \phi_ru^2 dr \quad (33)$$

We will estimate the terms on the right hand side. Applying the Cauchy-Schwartz inequality, we have

$$\int_0^R \frac{1}{r}uu_r dr \leq \int_0^R \frac{2u^2}{r^2} dr + \int_0^R \frac{1}{8}u_r^2 dr \quad \text{and} \quad \int_0^R u_r\phi_r dr \leq \int_0^R \frac{1}{2p}u_r^2 + \frac{p}{2}\phi_r^2 dr \quad (34)$$

and

$$\int_0^R u\phi dr \leq \int_0^R \frac{1}{2s}u^2 + \frac{s}{2}\phi^2 dr \quad (35)$$

Using Hardy's inequality, and assuming that $\phi(0) = 0$. (We will construct ϕ so that it will satisfy $\phi(0) = 0$.) We have

$$\int_0^R \frac{1}{r}\phi u_r dr \leq \left(\int_0^R \left(\frac{\phi}{r}\right)^2 dr\right)^{1/2} \left(\int_0^R u_r^2 dr\right)^{1/2} \leq 2\|\phi_r\|_2 \|u_r\|_2 \leq \int_0^R q\phi_r^2 + \frac{1}{q}u_r^2 dr \quad (36)$$

In order to find an estimate for $\int_0^R \frac{1}{r^2}u\phi dr$, we will first use Cauchy-Schwartz inequality then Hardy's Inequality. (since $u(0)=0$)

$$\int_0^R \frac{1}{r^2}u\phi dr \leq \int_0^R \frac{1}{2m} \left(\frac{u}{r}\right)^2 + \frac{m}{2} \left(\frac{\phi}{r}\right)^2 dr \leq \int_0^R \frac{2}{m}u_r^2 + 2m\phi_r^2 dr \quad (37)$$

When we add all, the right hand side of (33) becomes

$$\leq \int_0^R \left(-\frac{7}{8} + \frac{1}{2p} + \frac{1}{q} + \frac{2}{m}\right)u_r^2 + \left(1 + \frac{1}{2s} - \phi_r\right)u^2 + \left(\frac{p}{2} + q + 2m\right)\phi_r^2 + \frac{s}{2}\phi^2 dr \quad (38)$$

If we take $p = 12, q = 24, m = 48, s = 1/2$, then we get

$$\frac{1}{2} \frac{d}{dt} \|u - \phi\|^2 \leq \int_0^R -\frac{3}{4} u_r^2 + (2 - \phi_r) u^2 dr + \int_0^R 126 \phi_r^2 + \frac{1}{4} \phi^2 dr \quad (39)$$

Remark 0.5 *The preceding lemmas show that if we can construct $\phi \in \dot{H}^1[0, R]$ such that the coercivity estimate*

$$\langle u, Ku \rangle = 2 \int_0^R \frac{3}{4} u_r^2 + (\phi_r - 2) u^2 dr \geq \lambda_0 \|u\|^2 \quad (40)$$

holds for some λ_0 independent of L , then we get

$$\limsup_{t \rightarrow \infty} \|u\|_2 \leq c \|\phi\|_{\dot{H}^1[0, R]} \quad (41)$$

One can observe that the coercivity equation for radial case (40) is same as the coercivity equation (12) for one dimensional case. Thus the function ϕ_r exists.

Also note that for dimension n , the only difference as compared to dimension 2, will be

$$\Delta \phi = \left(\frac{\partial^2}{\partial r^2} + \frac{n-1}{r} \frac{\partial}{\partial r} \right) \phi. \quad (42)$$

The steps in the proof for dimension n will also be similar. The coefficients of the (0.4) will change, however these changes will not affect the exponent of the bound.

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